

MORRIS, MN, USA

WMO: 726565

Lat: 45.566N

Lon: 95.968W

Elev: 1130

StdP: 14.11

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 04948

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-17.3	-11.4	-23.5	1.6	-17.2	-17.7	2.2	-11.1	35.4	10.9	32.5	9.6	8.4	320	0.653

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	20.7	89.8	73.8	86.0	71.9	82.5	69.9	77.5	85.3	75.3	82.4	73.3	80.0	14.2	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
75.0	137.0	82.0	72.8	127.0	79.6	71.7	122.1	78.8	41.8	84.5	39.6	82.4	37.7	80.5	85.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
30.2	26.5	23.9	DB	-21.9	94.0	6.3	3.0	-26.4	96.2	-30.1	97.9	-33.7	99.6	-38.3	101.8	(4)
			WB	-22.4	80.7	6.1	2.5	-26.8	82.5	-30.3	84.0	-33.8	85.4	-38.2	87.2	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	43.2	11.6	15.0	28.7	43.5	56.4	66.8	70.7	67.8	60.5	46.7	31.3	17.3
	DBStd	23.24	13.94	13.39	12.89	10.14	8.87	6.77	5.82	5.62	8.27	9.34	11.38	12.27
	HDD50	4886	1192	981	665	240	39	1	0	0	10	176	568	1014
	HDD65	8476	1657	1401	1124	645	294	57	17	36	182	572	1012	1479
	CDD50	2391	0	0	6	46	236	505	643	550	327	73	5	0
	CDD65	507	0	0	0	1	26	111	195	122	48	4	0	0
	CDH74	4915	0	0	2	38	363	1070	1816	1057	503	66	0	0
	CDH80	1454	0	0	0	10	118	329	582	265	134	16	0	0
Wind	WSAvg	10.6	11.6	11.6	11.8	12.3	12.1	10.1	8.0	7.7	9.2	10.9	11.1	11.2
Precipitation	PrecAvg	25.6	0.8	0.7	1.3	2.4	2.9	4.0	3.9	3.2	2.4	2.2	1.1	0.7
	PrecMax	34.1	2.7	1.7	3.8	6.2	6.7	7.6	9.1	7.4	6.0	9.2	3.7	3.5
	PrecMin	9.9	0.0	0.1	0.2	0.2	0.3	0.5	1.1	0.2	0.2	0.1	0.1	0.0
	PrecStd	5.6	0.6	0.4	0.8	1.4	1.5	1.9	1.9	1.7	1.3	1.9	0.9	0.6
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	43.3	49.6	66.1	79.3	90.1	91.3	92.7	90.3	89.9	81.8	66.1	47.8
		MCWB	37.6	46.3	54.4	59.5	66.6	72.5	77.5	77.0	72.3	62.8	55.1	40.9
	2%	DB	36.7	41.4	57.5	72.1	82.4	88.0	89.7	85.7	83.7	73.1	58.7	39.3
		MCWB	34.3	38.0	50.5	55.5	64.0	71.3	76.5	74.4	68.9	58.7	49.6	35.6
	5%	DB	34.2	36.6	51.8	65.9	77.5	83.7	85.9	82.2	80.8	66.2	52.2	36.3
		MCWB	32.8	34.8	46.6	52.8	61.6	69.0	73.7	71.4	67.2	56.0	46.3	34.3
	10%	DB	30.3	34.2	45.4	61.2	72.5	80.8	82.4	80.8	75.2	62.5	46.0	33.9
		MCWB	29.7	32.9	42.0	50.2	58.9	66.9	71.8	70.3	64.4	53.3	41.6	32.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	38.5	45.4	57.3	62.3	71.1	77.1	81.3	79.5	75.2	66.6	57.5	41.7
		MCDB	42.3	47.6	63.0	75.6	82.7	86.7	87.3	86.2	83.5	76.3	64.4	46.3
	2%	WB	34.8	38.4	52.0	57.8	67.2	74.5	78.5	76.3	72.7	61.6	50.7	36.8
		MCDB	36.7	41.0	57.6	67.9	76.9	83.4	85.9	83.1	80.5	70.1	57.2	39.1
	5%	WB	32.5	35.3	46.3	54.2	64.1	71.9	76.3	74.1	69.9	57.3	46.2	34.3
		MCDB	33.9	36.4	50.0	63.6	73.0	80.0	83.5	80.1	76.0	64.7	51.4	36.0
	10%	WB	29.9	32.6	41.7	51.0	61.0	69.3	74.2	71.9	66.9	54.1	42.0	31.9
		MCDB	30.6	34.0	45.9	59.4	70.2	76.8	80.7	78.0	73.7	60.4	46.4	33.5
Mean Daily Temperature Range	5% DB	MDBR	16.8	17.9	17.2	20.7	21.8	20.5	20.7	21.2	22.8	19.9	17.2	15.7
		MCDBR	18.6	19.0	25.7	31.1	29.3	25.4	23.5	23.7	27.8	29.1	26.0	20.1
		MCWBR	16.3	16.2	17.9	17.4	14.9	11.9	12.3	13.1	14.2	17.4	17.5	16.4
	5% WB	MCDBR	17.5	17.3	24.0	27.5	24.3	20.9	21.0	21.1	22.3	25.1	24.4	18.5
	MCWBR	15.5	14.9	17.4	16.9	14.5	12.0	12.9	13.5	13.6	16.3	17.1	15.5	
Clear-Sky Solar Irradiance	taub	0.268	0.282	0.308	0.358	0.381	0.387	0.388	0.393	0.359	0.312	0.281	0.261	
	taud	2.415	2.386	2.431	2.344	2.343	2.378	2.363	2.335	2.418	2.533	2.530	2.454	
	Ebn at Noon	270	287	294	286	281	279	277	271	272	274	266	262	
	Edh at Noon	22	28	31	37	38	37	37	37	31	24	20	19	
All-Sky Solar Radiation	RadAvg	549	895	1207	1516	1733	1935	2056	1718	1316	832	535	390	
	RadStd	65	99	109	121	127	120	77	144	94	115	55	45	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only			N/A	N/A	N/A	N/A	N/A	N/A	N/A	+252	N/A
Regional (45 neighbors)			N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page